

Related Work

This review examines five studies directly relevant to AI-based Alzheimer's disease (AD) diagnosis from MRI. It spans a systematic review and meta-analysis of MCI-to-AD conversion models, single-model deep learning pipelines (Inceptionv3+DEMNET, hybrid 3D-CNN-Transformer), a comparative evaluation of CNN, BayesianCNN, and U-Net architectures, and a comprehensive survey of the field's translational challenges. For each study, the extracted information below covers dataset, application area, method, metrics, best results, strengths, limitations, research gap, and relation to the proposed work, followed by a consolidated related-work table.

1. Deep learning models predicting MCI-to-Alzheimer's conversion via structural MRI: A systematic review and meta-analysis

Authors: Kiani, Hajiasadi, Tahamtani Torkamani, Bagherzadeh, Abbasifar, Khademi Zare, Majidzade Moqadam, Mohammadzadeh, Kakadiya, Khosravi & Aghamollaii

Publication Year: 2026

Dataset(s) Used: 42 pooled studies, most using ADNI (alone or combined with other cohorts)

Application Area: Prediction of MCI-to-Alzheimer's disease conversion from structural MRI

Method / Model Used: Systematic review and meta-analysis of deep learning models (mostly CNNs, a growing subset transformer-based/hybrid); bivariate random-effects pooling; PRISMA 2020 methodology; METRICS quality tool

Performance Metrics: Sensitivity, specificity, AUC, diagnostic odds ratio (DOR), I^2 heterogeneity

Best Reported Results: Pooled sensitivity 76% (95% CI 72-79%); specificity 79% (95% CI 76-82%); DOR 11.63 (95% CI 8.86-15.27), attenuated to 7.55 after publication-bias adjustment. 2D models: sensitivity 84%, specificity 83%. Transfer learning: sensitivity 79% vs. 74% trained-from-scratch

Strengths: Large-scale quantitative synthesis across 42 studies; rigorous PRISMA-based protocol with formal quality appraisal; identifies concrete sources of heterogeneity (input dimensionality, training strategy, architecture)

Limitations: Very high between-study heterogeneity (generalized $I^2 > 84\%$); heavy reliance on ADNI limits generalizability; data leakage and absence of nested cross-validation in many included studies; near-absence of prospective/external validation; likely publication bias

Research Gap: No standardized methodological frameworks, prospective multi-site validation, or transparent/reproducible reporting to bridge research-stage performance and real clinical utility

Relation to Proposed Work: Sets the current pooled performance ceiling (~76-79% sensitivity/specificity) and exposes reproducibility and generalization weaknesses (ADNI dependence, data leakage) that the proposed work should explicitly correct through external, cross-dataset validation.

2. Early diagnosis of Alzheimer's disease from MRI images with deep learning model

Authors: Aghasi Javid & Mohassel Feghhi

Publication Year: 2024

Dataset(s) Used: Kaggle Alzheimer's 4-class MRI dataset (6,400 images; balanced to 12,800 via SMOTE)

Application Area: Multi-class AD stage classification (Mild / Moderate / Non- / Very Mild Demented)

Method / Model Used: Hybrid model: pre-trained Inceptionv3 for feature extraction + DEMNET CNN for classification; SMOTE for class-imbalance correction; RMSprop optimizer

Performance Metrics: Accuracy, precision, recall, F1-score

Best Reported Results: 98.67% overall accuracy; per-class precision/recall/F1 between 0.96 and 1.00

Strengths: Effectively resolves severe class imbalance (64 to 3,200 images per class) via SMOTE; combines a strong pretrained feature extractor with a lightweight dementia-specific classifier; very high reported accuracy

Limitations: Relies on an uncured Kaggle dataset with no documented acquisition or labeling protocol; single-dataset evaluation with no external/clinical validation; synthetic oversampling raises overfitting risk; no computational-efficiency analysis

Research Gap: No validation on curated clinical datasets (e.g., ADNI) and no evidence of cross-dataset generalizability

Relation to Proposed Work: Demonstrates a transfer-learning-plus-oversampling pipeline effective for imbalanced MRI classification, informing preprocessing and imbalance-handling choices while underscoring the need for clinically validated data in the proposed work.

3. Deep Learning for Early Alzheimer Disease Detection with MRI Scans

Authors: Rafsan, Oraby, Roy, Kumar & Rodrigo

Publication Year: 2025

Dataset(s) Used: OASIS brain MRI dataset (11,655 images from 1,378 subjects); Kaggle multiclass MRI dataset for comparison

Application Area: Multi-class dementia stage classification (Non-, Very Mild, Mild, Moderate Demented)

Method / Model Used: Comparative evaluation of ADD-Net (CNN), Bayesian CNN, and U-Net, combined with SMOTE-Tomek balancing and Grad-CAM interpretability

Performance Metrics: Accuracy, precision, recall, F1-score, AUC

Best Reported Results: BayesianCNN + SMOTE-Tomek: 95.03% accuracy on OASIS (97.49% on Kaggle data, F1 96.09%); ADD-Net: 94.16%; U-Net: 84.91%

Strengths: Direct head-to-head comparison of three DL paradigms on the same dataset; BayesianCNN adds uncertainty quantification relevant to clinical risk; Grad-CAM used for interpretability across all models

Limitations: U-Net omitted manual lesion masking, risking learning of irrelevant features; BayesianCNN mainly tuned on Kaggle data and needs further OASIS-specific adaptation; no demographic-stratified analysis; added computational cost of Bayesian inference

Research Gap: Lack of demographic-aware evaluation and limited exploration of advanced masking or attention-based U-Net variants for AD segmentation/classification

Relation to Proposed Work: Confirms Bayesian uncertainty quantification and interpretability (Grad-CAM) as valuable additions for clinical decision support, supporting their consideration in the proposed work's model architecture.

4. Advances in AI-based diagnosis of Alzheimer's disease using MRI: a comprehensive survey

Authors: Alyaqoobi, Lopez-Guede, Dara, Ramos-Hernanz, Aramendia & Teso-Fz-Betoño

Publication Year: 2026

Dataset(s) Used: Narrative synthesis of studies (2000-2025) predominantly using ADNI and OASIS

Application Area: Broad survey of AI/deep-learning-based Alzheimer's diagnosis using MRI

Method / Model Used: Comprehensive literature survey comparing CNNs, ResNet, DenseNet, ViT, LSTM, and classical ML (SVM, Random Forest, k-NN, ensembles)

Performance Metrics: Accuracy as reported by surveyed studies (ranges rather than pooled statistics)

Best Reported Results: Best individual study cited: ~98.82% accuracy (ensemble of three DenseNet architectures); classical ML accuracy generally 82-91%

Strengths: Broad 25-year synthesis of the field; distills five concrete translational barriers (data scarcity, overfitting, interpretability, computational cost, generalization); proposes actionable future directions (federated learning, XAI, standardized benchmarks)

Limitations: Narrative rather than quantitative/meta-analytic synthesis; no formal pooled effect sizes; relies on secondary reporting of accuracy figures without independent re-verification

Research Gap: Limited adoption of federated learning, explainable AI frameworks, and standardized benchmarking protocols needed for clinical translation

Relation to Proposed Work: Frames the field-level obstacles (data scarcity, interpretability, generalization) that the proposed work must design around, motivating inclusion of explainable AI and cross-dataset benchmarking.

5. A deep learning model for early diagnosis of alzheimer's disease combined with 3D CNN and video Swin transformer

Authors: Zhou, Wei, Li, Zhou, Tao, Hua & Liu**Publication Year:** 2025**Dataset(s) Used:** ADNI (798 3D MRI images, one scan per subject); external generalization test on OASIS (104 images)**Application Area:** Binary classification of Alzheimer's disease vs. cognitively normal (CN) from 3D whole-brain MRI

Title	Authors	Year	Dataset	Application Area	Method/Model	Metrics	Strengths	Limitations	Research Gap
Deep learning models predicting MCI-to-Alzheimer's conversion via structural MRI: A systematic review and meta-analysis	Kiani, Hajiasadi, Tahamtani, Torkamani, Bagherzadeh, Abbasifar, Khademi Zare, Majidzade Moqadam, Mohammadzadeh, Kakadiya, Khosravi & Aghamollaii	2026	42 pooled studies, most using ADNI (alone or combined with other cohorts)	Prediction of MCI-to-Alzheimer's disease conversion from structural MRI	Systematic review and meta-analysis of deep learning models (mostly CNNs, a growing subset transformer-based/hybrid); bivariate random-effects pooling; PRISMA 2020 methodology; METRICS quality tool	Sensitivity, specificity, AUC, diagnostic odds ratio (DOR), I ² heterogeneity	Large-scale quantitative synthesis across 42 studies; rigorous PRISMA-based protocol with formal quality appraisal; identifies concrete sources of heterogeneity (input dimensionality, training strategy, architecture)	Very high between-study heterogeneity (generalized I ² > 84%); heavy reliance on ADNI limits generalizability; data leakage and absence of nested cross-validation in many included studies; near-absence of prospective/external validation; likely publication bias	No standardized methodology frameworks, prospective multi-center validation, or transparent/reproducible reporting to bridge research-performance and real clinical utility
Early diagnosis of Alzheimer's disease from MRI images with deep learning model	Aghasi Javid & Mohassel Feghhi	2024	Kaggle Alzheimer's 4-class MRI dataset (6,400 images; balanced to 12,800 via SMOTE)	Multi-class AD stage classification (Mild / Moderate / Non- / Very Mild Demented)	Hybrid model: pre-trained Inceptionv3 for feature extraction + DEMNET CNN for classification; SMOTE for class-imbalance correction; RMSprop optimizer	Accuracy, precision, recall, F1-score	Effectively resolves severe class imbalance (64 to 3,200 images per class) via SMOTE; combines a strong pretrained feature extractor with a lightweight dementia-specific classifier; very high reported accuracy	Relies on an uncensored Kaggle dataset with no documented acquisition or labeling protocol; single-dataset evaluation with no external/clinical validation; synthetic oversampling raises overfitting risk; no computational-efficiency analysis	No validation on curated clinical datasets (e.g., ADNI) and no analysis of cross-dataset generalizability
Deep Learning for Early Alzheimer Disease Detection with MRI Scans	Rafsan, Oraby, Roy, Kumar & Rodrigo	2025	OASIS brain MRI dataset (11,655 images from 1,378 subjects); Kaggle multiclass MRI dataset for comparison	Multi-class dementia stage classification (Non-, Very Mild, Mild, Moderate Demented)	Comparative evaluation of ADD-Net (CNN), Bayesian CNN, and U-Net, combined with SMOTE-Tomek balancing and Grad-CAM interpretability	Accuracy, precision, recall, F1-score, AUC	Direct head-to-head comparison of three DL paradigms on the same dataset; BayesianCNN adds uncertainty quantification relevant to clinical risk; Grad-CAM used for interpretability across all models	U-Net omitted manual lesion masking, risking learning of irrelevant features; BayesianCNN mainly tuned on Kaggle data and needs further OASIS-specific adaptation; no demographic-stratified analysis; added computational cost of Bayesian inference	Lack of demographic-aware evaluation and limited exploration of advanced masking or attention mechanisms; need for U-Net variants for AD segmentation/classification
Advances in AI-based diagnosis of Alzheimer's disease using MRI: a comprehensive survey	Alyaqoobi, Lopez-Guede, Dara, Ramos-Hernanz, Aramendia & Teso-Fz-Betoño	2026	Narrative synthesis of studies (2000-2025) predominantly using ADNI and OASIS	Broad survey of AI/deep-learning-based Alzheimer's diagnosis using MRI	Comprehensive literature survey comparing CNNs, ResNet, DenseNet, ViT, LSTM, and classical ML (SVM, Random Forest, k-NN, ensembles)	Accuracy as reported by surveyed studies (ranges rather than pooled statistics)	Broad 25-year synthesis of the field; distills five concrete translational barriers (data scarcity, overfitting, interpretability, computational cost, generalization); proposes actionable future directions (federated	Narrative rather than quantitative/meta-analytic synthesis; no formal pooled effect sizes; relies on secondary reporting of accuracy figures without independent re-verification	Limited adoption of federated learning, explainable AI frameworks, and standardized benchmarking protocols needed for clinical translation

Title	Authors	Year	Dataset	Application Area	Method/Model	Metrics	Strengths	Limitations	Research Gap
							learning, XAI, standardized benchmarks)		
A deep learning model for early diagnosis of alzheimer's disease combined with 3D CNN and video Swin transformer	Zhou, Wei, Li, Zhou, Tao, Hua & Liu	2025	ADNI (798 3D MRI images, one scan per subject); external generalization test on OASIS (104 images)	Binary classification of Alzheimer's disease vs. cognitively normal (CN) from 3D whole-brain MRI	3D-CNN-VSwinFormer: 3D CNN with a 3D Convolutional Block Attention Module (CBAM) combined with a fine-tuned Video Swin Transformer	Accuracy, AUC, sensitivity, specificity, model parameters/FLOPs	Avoids 2D-slice data leakage by using one 3D scan per subject; fuses local (CNN) and global (Transformer) feature extraction; explicitly tests cross-dataset generalization; favorable parameter/FLOP efficiency versus comparably-accurate models	Small training set (798 images) constrains generalization; marked accuracy drop on external OASIS data (92.92% to 75.96%); relatively high computational cost (146.42 GFLOPs); dependence on a single primary data source (ADNI)	Insufficient large-scale, diverse MRI datasets and limited modeling of longitudinal/temporal disease progression

Method / Model Used: 3D-CNN-VSwinFormer: 3D CNN with a 3D Convolutional Block Attention Module (CBAM) combined with a fine-tuned Video Swin Transformer

Performance Metrics: Accuracy, AUC, sensitivity, specificity, model parameters/FLOPs

Best Reported Results: 92.92% accuracy and AUC 0.9660 (sensitivity 95.41%) on ADNI; 75.96% accuracy and AUC 0.8177 on external OASIS test, best among compared models

Strengths: Avoids 2D-slice data leakage by using one 3D scan per subject; fuses local (CNN) and global (Transformer) feature extraction; explicitly tests cross-dataset generalization; favorable parameter/FLOP efficiency versus comparably-accurate models

Limitations: Small training set (798 images) constrains generalization; marked accuracy drop on external OASIS data (92.92% to 75.96%); relatively high computational cost (146.42 GFLOPs); dependence on a single primary data source (ADNI)

Research Gap: Insufficient large-scale, diverse 3D MRI datasets and limited modeling of longitudinal/temporal disease progression

Relation to Proposed Work: Provides an architecture-level benchmark (hybrid CNN-Transformer) and highlights a clear cross-dataset generalization gap, motivating the proposed work's emphasis on multi-dataset validation and dataset diversity.

Related Work Table

Summary of Key Themes

Across these five studies, ADNI and OASIS remain the dominant data sources, and CNN-based architectures (plain, Bayesian, or hybridized with attention/Transformer modules) consistently outperform classical machine learning baselines, with reported accuracies ranging from the mid-80s to over 98% depending on dataset curation and task difficulty. The pooled meta-analytic evidence (Kiani et al., 2026), however, tempers these single-study figures: once heterogeneity, data leakage, and publication bias are accounted for, realistic sensitivity and specificity settle closer to 76-79%. A recurring methodological theme is heavy dependence on a single dataset — whether Kaggle, OASIS, or ADNI — with limited cross-dataset testing; where such testing was attempted (Zhou et al., 2025), accuracy dropped substantially when moving from ADNI to OASIS, exposing a generalization gap. Interpretability (Grad-CAM, Bayesian uncertainty) and class-imbalance handling (SMOTE, SMOTE-Tomek) emerge as consistent design elements needed to make results clinically credible. The survey by Alyaqoobi et al. (2026) ties these threads together, identifying data scarcity, overfitting, interpretability, computational cost, and generalization as the field's core unresolved barriers. Collectively, this body of work points to a clear gap that the proposed research aims to address: a deep learning approach for AD diagnosis from MRI that is validated across multiple independent datasets, incorporates interpretability and uncertainty estimation, and is benchmarked using standardized, leakage-free protocols rather than single-dataset accuracy alone.